DDIN TC AGENT VISIT & REPORTING SYSTEM

1. What This System Does

This system turns paper-based field visits into a digital process. Territory Coordinators (TCs) visit DDIN agents in person, and every visit is recorded in the system. Each visit record links together the TC who made the visit, the agent who was visited, the agent's registered location, the TC's real GPS location at the time, the date and time, a written report, and a photo taken during the visit.

2. The Basic Rule

A TC must actually be present at the agent's registered location before a visit can be started or completed as "verified". The system captures GPS coordinates automatically and calculates the distance between the TC and the agent — the TC cannot type this in manually or fake it.

3. Who Uses the System

There are three main types of users:

- TCs (Territory Coordinators) — carry out visits and submit reports.
- Champions / supervisors — monitor TCs, agents, and visits.
- Management / administrators — manage master data, assignments, settings, dashboards, and reports.

4. Agent Registration

Every agent in the system needs the following information on file:

- A unique Agent ID, name, and phone number
- Business or address details
- Province, district, sector, and territory
- The TC assigned to that agent
- Status (active, inactive, etc.)
- Registered GPS coordinates (latitude and longitude)
- GPS accuracy and location-verification status

5. TC Assignment

Every active TC has a login account and a list of agents they are authorized to visit. When assignments change, the system must keep a history — including when each assignment started and ended — rather than simply overwriting the old assignment.

6. How a Visit Works, Step by Step

1. The TC logs in.
2. The TC selects an assigned agent.
3. The TC views that agent's registered location.
4. The TC travels to the agent.
5. The TC requests to check in.
6. The system captures the TC's GPS location automatically.
7. The backend calculates the distance between the TC and the agent.
8. The location is marked as verified (or not).
9. The TC takes the required photo.
10. The TC completes the visit report.
11. The TC checks out.
12. The system validates everything and stores the completed visit.

7. Location Check (GPS and Geofence)

The system compares the TC's current GPS position with the agent's registered location. The recommended starting radius is 100 metres, and this distance should be adjustable by an administrator later. The system stores both the calculated distance and the GPS accuracy. If a TC is outside the allowed radius, they cannot submit a visit as "verified".

8. Visit Photo

Every completed visit must include a photo. The photo must:

- Be taken from within the visit workflow itself (not uploaded from the phone's gallery)
- Be linked to that specific visit
- Be time-stamped
- Be stored securely
- Be protected so it cannot be swapped or replaced without a record of the change

DDIN still needs to decide exactly what the photo should show (for example, the agent's storefront, signage, or the agent in person).

9. Visit Report

Each visit report should capture:

- The agent's operating status

- Services and products offered
- General activity observations
- Liquidity or cash-float observations, where relevant
- Customer or service-quality observations
- Challenges the agent is facing
- Support needs and training needs
- Follow-up actions required
- Any additional remarks from the TC

The exact list of questions (the checklist) should be configurable, so DDIN can adjust it without needing new software development.

10. Making Sure Visits Are Genuine

Before accepting a visit as valid, the system checks the TC's identity, whether the TC is authorized for that agent, GPS proximity, GPS accuracy, timestamps, whether the required photo exists, and whether the visit was actually completed. Visits that look unusual should be flagged for a human to review — the system should not automatically brand them as misconduct.

11. What Happens When Something Goes Wrong

- TC is outside the geofence: the system blocks a verified check-in.
- GPS signal is weak: the system warns the TC or flags the visit.
- GPS is unavailable: the visit cannot be location-verified.
- The agent has physically moved: use the official location-update process to fix the agent's registered location.
- The photo is missing: the visit cannot be marked complete.
- No internet connection: the TC can still capture the visit offline, and it syncs to the server once back online.

12. Reporting

Reports must be filterable by date, TC, Champion/supervisor, territory, province, district, sector, and agent. Core reports should include: TC visits, agent visit history, coverage, verified vs. flagged visits, failed location verifications, and visit evidence (photos and reports).

13. Key Numbers to Track

- Visit Coverage = Agents Visited ÷ Agents Due × 100

- Verified Visit Rate = Verified Visits ÷ Completed Visits × 100
- Flagged Visit Rate = Flagged Visits ÷ Completed Visits × 100
- Photo Evidence Rate = Visits With a Valid Photo ÷ Completed Visits × 100

14. How the Data Connects

The core chain of information is: TC → Visit → Agent → Registered Agent Location. Every visit record keeps a permanent link to the TC, the agent, the territory, the agent's registered location, the TC's actual location, the time, the report, and the photo.

15. Database

The recommended core tables are:

- users
- territories
- agents
- agent_locations
- tc_agent_assignments
- visits
- visit_reports
- visit_photos
- issues
- follow_ups
- audit_logs

PostgreSQL with the PostGIS extension is recommended, since it handles geographic (location-based) verification well.

16. Security and Privacy

The system should use:

- Role-based access (each user only sees what their role allows)
- HTTPS for all connections
- Secure authentication
- Server-side authorization checks (not just checks on the phone app)
- Protected photo storage
- Audit logs and regular backups

By default, a TC's location should only be collected for the purpose of the visit itself — not tracked continuously throughout the day. Access to location data and photos must be restricted to authorized roles.

17. What "Done" Looks Like

- An agent can be registered with a location.
- A TC can see and access their authorized agents.
- GPS is captured automatically, without manual entry.
- The backend calculates the distance between TC and agent.
- Visits started outside the allowed radius cannot be marked as location-verified.
- A required photo is attached to every completed visit.
- Every visit links exactly one TC and one agent.
- Timestamps, GPS coordinates, distance, and accuracy are all stored.
- Management can filter and report on visits.
- Historical assignments and past locations are preserved, not overwritten.

18. What to Build First (MVP)

The first version should include:

- Authentication (login)
- Agent registration
- Registered GPS for each agent
- TC assignments
- Mobile visit workflow
- GPS / geofence verification
- Check-in and check-out
- Mandatory photo capture
- Structured visit report
- Visit history
- Basic reporting
- Audit trail

19. Main Visit Record — What Gets Stored

Each visit record contains the following fields:

Field	What It Means	Where It Comes From
visit_id	Unique ID for the visit	System
agent_id	Which agent was visited	System
tc_id	Which TC conducted the visit	Logged-in user
territory_id	Territory at the time of the visit	System
visit_date	Date of the visit	System
check_in_time	When the visit started	System
check_out_time	When the visit finished	System
agent_latitude / agent_longitude	Agent's registered location	Agent record
tc_latitude / tc_longitude	TC's location during the visit	Device GPS
distance_meters	Distance between TC and agent	Calculated by backend
gps_accuracy_meters	How accurate the GPS reading was	Device GPS
location_status	VERIFIED / OUTSIDE_RADIUS / LOW_ACCURACY	System
photo_id	The required visit photo	System
photo_captured_at	When the photo was taken	System
visit_status	STARTED / COMPLETED / FLAGGED / REJECTED	System
report_data	The TC's structured answers	TC
remarks	Any extra notes	TC
created_at / updated_at	Record timestamps	Server

20. A Simple Example, Start to Finish

1. TC-023 selects Agent AGT-00245 from their list.
2. The system shows the agent's registered location.
3. TC-023 travels there and requests check-in.
4. The device captures the TC's GPS position and accuracy.
5. The backend calculates the distance to the agent's registered location.
6. The distance comes out to 32 metres, and the configured radius is 100 metres.
7. The location status becomes VERIFIED.
8. TC-023 takes the required visit photo.
9. TC-023 completes the visit report and checks out.
10. The system stores the visit, GPS data, timestamps, distance, report, and photo.
11. Later, management can see exactly which TC visited which agent, when, where, and with what evidence.

21. Organizational Structure

The reporting structure for this specification is:

District → Territory Coordinator (TC) → Sector → Champion → Agent

A TC operates at district level and supervises Champions. One TC may be responsible for more than one district. Champions are assigned at sector level and report to their TC.

Level	Role	Coverage	Access
District	Territory Coordinator (TC)	One or more districts	Assigned districts, sectors, Champions, agents, and visits
Sector	Champion	One sector	Agents and visits within the assigned sector
Agent	Agent	Registered agent location	Own profile and visit-related information

22. Champion and TC Access Rules

Access follows organizational responsibility:

- A TC's access is based on their assigned district(s).
- A Champion's access is based on their assigned sector.
- A TC can view and monitor Champions within their assigned districts.
- A TC can view agents and visits across all sectors under their assigned districts.
- One TC account can be responsible for more than one district.
- A Champion can view agents, visits, photos, reports, issues, and follow-ups within their assigned sector.
- A Champion cannot access another sector unless explicitly authorized.
- Management/Admin can access all areas, or a configured subset of areas.
- Access and important management actions must be recorded in the audit trail.

23. Access Matrix

Role	Level	Agents	Visits / Photos	Reports
Management / Admin	National / authorized areas	All authorized	All authorized	National and detailed
TC	One or more districts	Agents under assigned districts	View/manage according to permissions	District and sector
Champion	Assigned sector	Agents in sector	View/review sector visits and photos	Sector

Role	Level	Agents	Visits / Photos	Reports
Agent	Own record	Own profile	Own visit history where enabled	Own information

24. Overall Field Visit Model

The system's main purpose is to record and verify physical visits made by TCs to DDIN agents. TCs are the primary field users. Every visit must be linked to the TC, the agent, the agent's registered location, the TC's actual GPS location, the visit date and time, photo evidence, and the visit report.

Organizational structure

District → TC → Sector → Champion → Agent

Visit flow

TC → physically visits → Agent → GPS verification → Photo → Visit Report → Submission

25. TC Field Visit Requirement

TCs are responsible for physically visiting agents within their assigned district(s). One TC may be responsible for more than one district, so the system must allow a single TC account to hold multiple district assignments and must show that TC all agents authorized under those districts.

- TC selects the agent to be visited.
- TC travels to the registered agent location.
- The system captures the TC's actual GPS position.
- The system compares the TC's position with the agent's registered location.
- Only an authorized, location-verified visit can proceed as a verified visit.
- TC captures a photo during every completed visit.
- TC completes the visit report.
- The system stores the complete visit record for reporting and audit.

26. What Every Visit Must Prove

Question	System Evidence
Who visited?	Authenticated TC ID
Who was visited?	Agent ID
Where was the agent registered?	Registered agent GPS coordinates
Where was the TC?	TC GPS coordinates captured during the visit

Question	System Evidence
How far apart were they?	Distance calculated by the backend
When did the visit occur?	System timestamps
What proves physical presence?	Required photo plus location verification
What happened during the visit?	Structured visit report

27. Role Responsibilities Summary

Role	Level	Primary Responsibility	Visit Access
Management / Admin	National / authorized	Manage and monitor the system and overall network	View authorized visits, photos, reports, and analytics
TC	District; may cover multiple districts	Primary field visitor; visits agents and supervises Champions	Create visits, GPS verification, photos, reports; view assigned districts
Champion	Sector	Supervises agents and field activity in the assigned sector; reports to TC	Review sector agents, visits, photos, reports, and follow-ups
Agent	Registered agent	Maintains agent identity/location and receives field visits	Own profile and visit history; confirmation if enabled